Extra Practice 1: Similar Polygons and Scale Factor

A. Skills

1. $\triangle RST \sim \triangle GHJ$. Complete the table, then write the three equal ratios of corresponding sides (new over original).

In $\triangle RST$	$\angle R$	$\angle S$	$\angle T$
In $\triangle GHJ$			

In $\triangle RST$	RS	ST	TR
In $\triangle GHJ$			

2. Find the scale factor k. Write enlargement, reduction or same size.

	Original	**New**	**Scale factor k**	**Type**
(a)	6 cm	27 cm		
(b)	25 m	10 m		
(c)	13 ft	13 ft		
(d)	8 cm	2 cm		
(e)	1.5 in	4.5 in		

3. A quadrilateral has sides 4 cm, 6 cm, 7 cm and 10 cm. Find the sides of a similar quadrilateral for each scale factor.

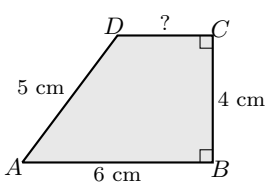
Original sides	4 cm	6 cm	7 cm	10 cm
(a) $k = 1.5$				
(b) $k = 0.5$				

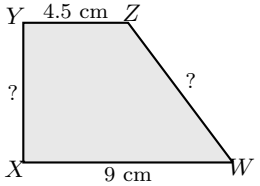
4. Quadrilateral $EFGH \sim$ quadrilateral $MNOP$. $\angle E = 95°$, $\angle F = 70°$ and $\angle G = 120°$. Find $\angle H$, then all four angles of $MNOP$.

B. Applying

5. $ABCD \sim WXYZ$. The diagrams are not drawn to scale.

(a) Find the scale factor from $ABCD$ to $WXYZ$.





(b) Find XY, ZW and CD.

6. A club logo is printed 8 cm wide and 5.6 cm tall on a T-shirt. A similar logo is printed on a banner. The banner logo is 2 m wide.

(a) Find the scale factor from the T-shirt to the banner.

(b) How tall is the banner logo?

7. A model train set uses the $1:87$ scale.
(a) A real locomotive is 22.62 m long. How long is the model, in centimetres?

(b) The model locomotive is 4.6 cm tall. How tall is the real one, to the nearest tenth of a metre?

8. A video is 12 cm wide and 5.4 cm tall on a phone. The same video fills a TV screen that is 108 cm wide. How tall is the video on the TV?

9. Sam changes an 8 cm by 12 cm photo to 16 cm by 20 cm. Is the new photo similar to the original? Explain. If not, what should the long side be?

10. $\triangle ABC \sim \triangle DEF$, and the scale factor from $\triangle ABC$ to $\triangle DEF$ is 3. If $AB = 4$ cm, then DE is
A. 1.3 cm **B.** 7 cm **C.** 12 cm **D.** 16 cm

11. Numerical Response A poster is 90 cm by 60 cm. It is reduced to a similar flyer whose short side is 21 cm. The long side of the flyer, to the nearest tenth of a centimetre, is _____ cm.

C. Stretch

12. A rectangle is 9 cm by 6 cm. It is enlarged with a scale factor of 1.8.
(a) Find the length and width of the new rectangle.

(b) Find both perimeters. How many times bigger is the new perimeter?

(c) Find both areas. How many times bigger is the new area? How is this number related to k?

13. A drawing is enlarged with a scale factor of 2. Then the copy is reduced with a scale factor of 0.75.
(a) A side of the first drawing is 8 cm. How long is it on the copy? On the final drawing?

(b) What single scale factor changes the first drawing into the final drawing? How could you find it from 2 and 0.75?

14. Figure B is an enlargement of figure A with a scale factor of 4.

(a) What scale factor changes figure B back into figure A?

(b) A side of figure B is 26 cm. How long is the matching side of figure A?

Extra Practice 2: Are They Similar? Checking Angles and Side Ratios

A. Skills

1. Quadrilaterals $ABCD$ and $EFGH$ have equal corresponding angles. Their sides are $AB = 5$, $BC = 8$, $CD = 6$, $DA = 4$ and $EF = 2$, $FG = 3.2$, $GH = 2.4$, $HE = 1.6$ (all in cm).

Pair	$EFGH \div ABCD$	**Ratio**
EF and AB		
FG and BC		
GH and CD		
HE and DA		

Similar? _____ Scale factor: _____ Enlargement or reduction? _____

2. Quadrilateral $PQRS$ has $\angle P = 85°$, $\angle Q = 95°$, $\angle R = 110°$. Quadrilateral $WXYZ$ has $\angle W = 85°$, $\angle Y = 110°$, $\angle Z = 70°$.

(a) Find $\angle S$ and $\angle X$.

(b) Does the angle test pass? Can you say the quadrilaterals are similar? Explain.

3. Which rectangles are similar to a 10 cm by 25 cm rectangle? $(25 \div 10 = 2.5)$

Rectangle	**Long ÷ short**	**Similar?**
(a) 4 cm by 10 cm		
(b) 20 cm by 40 cm		
(c) 75 cm by 30 cm		
(d) 12 cm by 27 cm		

4. Write *always*, *sometimes* or *never*.

(a) Two squares are similar. _____

(b) Two parallelograms are similar. _____

(c) Two regular octagons are similar. _____

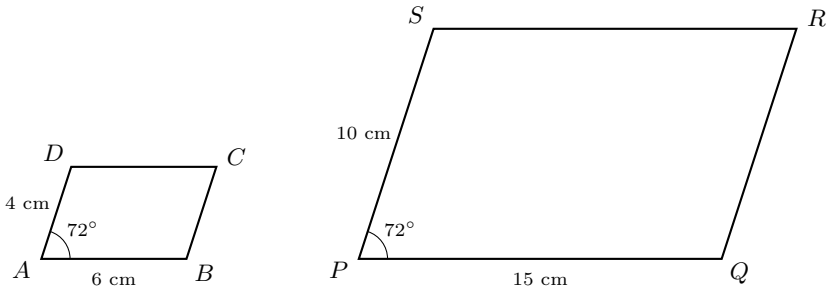
(d) An equilateral triangle and a square are similar. _____

B. Applying

5. One phone screen is 6.5 cm by 14 cm. Another is 7 cm by 15 cm. Are the screens similar? Show your work.

6. An 8 in by 10 in photo is framed with a 1.5 in border on every side. Is the framed photo similar to the photo? Show your work.

7. Are these parallelograms similar? (Not drawn to scale.)



8. A small candle is a cylinder with a radius of 3 cm and a height of 8 cm. A large candle has a radius of 4.5 cm and a height of 12 cm. Are the candles similar? Show your work.

9. Jess says a square with 5 cm sides and a rectangle that is 5 cm by 7 cm are similar, "because all their angles are 90°." Is Jess right? Explain.

10. A rectangle is 12 cm by 16 cm. Which rectangle is **not** similar to it?

A. 3 cm by 4 cm **B.** 18 cm by 24 cm **C.** 15 cm by 19 cm **D.** 30 cm by 40 cm

11. Numerical Response A rectangle is 15 cm by 24 cm. A similar rectangle has a short side of 10 cm. The length of its long side, in centimetres, is _____.

C. Stretch

12. Two hexagons have side ratios that are all 1.2. Four pairs of corresponding angles are equal: 110°, 130°, 125° and 120°. The other two angles are not given. Zoe says the hexagons must be similar. Is she right? Explain, and give an example.

13. Rectangle A is 3 cm by 5 cm. Rectangle B is 6 cm by 10 cm.

(a) Show that the rectangles are similar, and find the scale factor.

(b) Find the area of each rectangle. How many times bigger is the area of B?

(c) Rectangle C is 9 cm by 15 cm. Find its scale factor from A and how many times bigger its area is. What pattern

do you see?

14. A painting is 30 cm by 45 cm. It gets a border of the same width on every side.

(a) Find the outside size, and long ÷ short, for a 5 cm border and for a 10 cm border.

(b) The painting's ratio is $45 \div 30 = 1.5$. Can any border of the same width all around make the framed painting similar to the painting? Explain.

15. Describe (or sketch) each shape.

(a) A quadrilateral with the same angles as a 4 cm by 6 cm rectangle, but **not** similar to it.

(b) A quadrilateral whose sides are all in the same ratio to the sides of a square with 3 cm sides, but **not** similar to it.

Extra Practice 3: Drawing Enlargements and Reductions

A. Skills

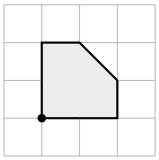
1. A quadrilateral has sides of 8 cm, 5 cm, 6 cm and 12 cm, and one angle of 110°. Complete the table for each similar copy.

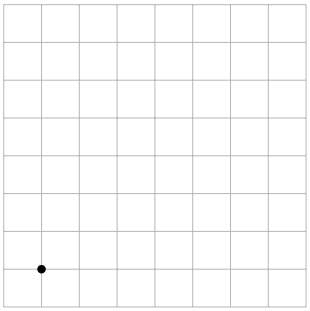
k	**Enlargement or reduction?**	**8 cm**	**5 cm**	**6 cm**	**12 cm**	**The 110° angle**
2.5						
0.25						
1.2						

2. Each copy was made from an original with the scale factor shown. Find the length of the matching side on the original.

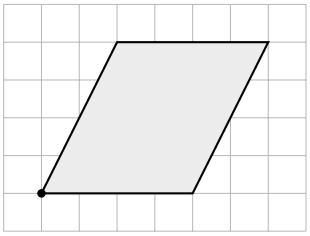
	(a)	(b)	(c)
Scale factor k	4	0.5	1.5
Side on the copy	26 cm	7 cm	12 cm
Side on the original			

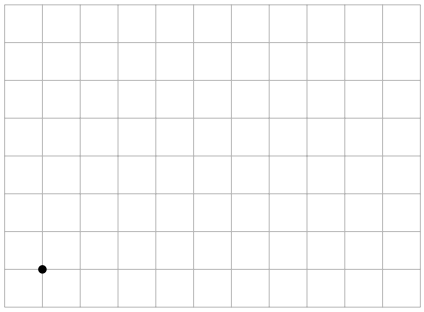
3. Draw an enlargement of the shape with $k = 3$. Start at the dot.



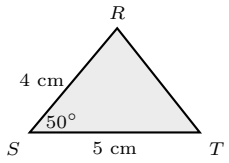


4. Draw an enlargement of the parallelogram with $k = 1.5$. Start at the dot.

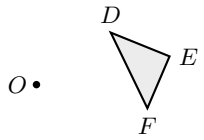




5. The sketch of $\triangle RST$ is not to scale. Use a ruler and protractor to draw an enlargement with $k = 1.2$ in the space. Then measure $R'T'$.



6. Draw an enlargement of $\triangle DEF$ with centre O and $k = 3$. Write each distance from O and each new distance.



B. Applying

7. A shipping container is 12 m long, 2.4 m wide and 2.6 m tall. A toy container is similar to it, with a scale factor of $\frac{1}{50}$. Find the toy's dimensions in centimetres.

8. The front of a garden shed is a pentagon: a 2.4 m base, two 2.0 m walls and two 1.5 m roof edges. Hana builds a similar model whose base is 12 cm.

(a) Find the scale factor. (Use the same units.)

(b) Find the model's wall and roof-edge lengths.

(c) The real roof has a 106° angle at the peak. What is the angle on the model?

9. Tariq wants to enlarge an 8 cm by 12 cm picture onto a 56 cm by 90 cm piece of poster board. The enlargement must be similar and fit on the board. Find the largest scale factor he can use, the size of the poster, and how much board is left over.

10. Noah "enlarges" a 2 cm by 3 cm rectangle with $k = 2$. He draws a 4 cm by 6 cm shape, but his corners are 80° and 100°. Is his shape similar to the rectangle? Explain.

11. Which of these does **not** make a figure similar to the original?

A. Multiplying every side by 2 and keeping every angle the same

B. The projection method with $k = 0.5$

C. Adding 2 cm to every side

D. On a grid, multiplying every move by 3

12. Numerical Response A side of a triangle is 7.2 cm. The triangle is reduced with $k = 0.25$. The length of the matching side, in centimetres, is _____ cm.

C. Stretch

13. A figure has a side of 12 cm. It is enlarged with $k = 2$. Then the enlargement is reduced with $k = \frac{1}{4}$.

(a) How long is the side in the final figure?

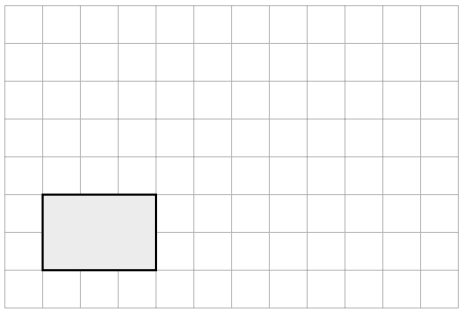
(b) What single scale factor takes the original straight to the final figure?

14. The grid shows a 2 by 3 rectangle.

(a) Draw an enlargement with $k = 2$ on the grid.

(b) How many squares are inside each rectangle? How many times bigger is the area of the enlargement?

(c) Did you know? The area grows by $k \times k$. Predict how many times bigger the area is when $k = 3$.



Extra Practice 4: Scale Diagrams: Maps and Floor Plans

A. Skills

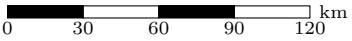
1. Complete the table.

	Scale in words	**As a ratio**	**Reduction or enlargement?**
(a)	1 cm represents 8 m		
(b)	1 cm represents 0.5 km		
(c)	1 in represents 6 ft		
(d)		1 : 40	
(e)		1 : 2 500 000	
(f)		10 : 1	

2. A plan uses the scale 1 cm represents 3 m. Find each real length.
(a) 4 cm _____ (b) 2.5 cm _____ (c) 0.6 cm _____ (d) 6.8 cm _____

3. A plan has a scale of 1 : 250. Find each plan length in centimetres.
(a) 10 m _____ (b) 6.5 m _____ (c) 1.5 m _____ (d) 22 m _____

4. A highway map has this bar scale. Use a ruler.



(a) Write the scale in words.

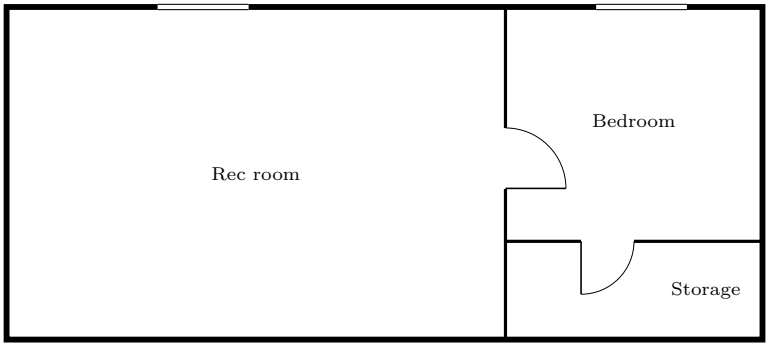
(b) Write the scale as a ratio.

(c) Two towns are 3.5 cm apart on this map. How far apart are they?

5. For each scale, is the drawing smaller than, bigger than, or the same size as the real object?
(a) 1 : 30 _____ (b) 4 : 1 _____ (c) 1 : 1 _____ (d) 1 : 100 000 _____

B. Applying

6. This basement plan is drawn to scale at 1 : 100. Use a ruler.



(a) Find the real length and width of the rec room.

(b) Find the real length and width of the bedroom.

(c) Carpet for the rec room costs $32.95 per square metre. Find the cost of carpet for the whole rec room floor, before tax.

7. A road map of Alberta has a scale of 1 : 2 500 000.

(a) Calgary and Lethbridge are 7 cm apart on the map. About how far apart are they (straight line)?

(b) Grande Prairie and Edmonton are about 390 km apart (straight line). How far apart are they on the map?

8. A die-cast model motorcycle is built at a scale of 1 : 18.

(a) The model is 12 cm long. How long is the real motorcycle, to the nearest tenth of a metre?

(b) The real motorcycle is 1.1 m tall. How tall is the model, to the nearest tenth of a centimetre?

9. Mr. Okafor wants to draw the school gym, 36 m long and 20 m wide, in a space 16 cm long and 12 cm wide.

(a) Choose an easy scale so the gym fits. Write it in words and as a ratio.

(b) Find the length and width of the gym on the drawing.

10. A plan for a garden shed uses the scale 1 in represents 3 ft.

(a) The shed is 12 ft long and 9 ft wide. Find its length and width on the plan.

(b) The door is $1\frac{1}{4}$ in wide on the plan. How wide is the real door, in feet and inches?

11. A room is 20 m long. Which scale gives the **longest** drawing of the room?

A. 1 : 50 **B.** 1 : 100 **C.** 1 : 200 **D.** 1 : 500

12. Numerical Response A map has a scale of 1 : 30 000. A bike route has three straight parts measuring 3.4 cm, 2.8 cm and 4.3 cm on the map. The real length of the route, in kilometres, is _____.

C. Stretch

13. Plan A of a house is drawn at 1 : 50. Plan B of the same house is drawn at 1 : 200. A living room is 5 m long.

(a) How long is the living room on each plan?

(b) How many times longer is every length on Plan A than on Plan B? Explain using the scales.

14. On a 1 : 100 floor plan, a bedroom is 4 cm by 3.5 cm.

(a) Meg finds the plan area, $4 \times 3.5 = 14$ cm^2, and multiplies by 100 to get 1400 cm^2, or 0.14 m^2. Why is her answer not reasonable?

(b) Find the real area correctly.

(c) The real area is how many times the plan area? (Hint: 1 m^2 = 10 000 cm^2.)

15. Two maps show the same provincial park. Map A has a scale of 1 : 25 000. Map B has a scale of 1 : 10 000.

(a) A trail is 8 cm long on Map A. How long is the real trail, in kilometres?

(b) How long is the same trail on Map B?

(c) Which map would you use to find a campsite in a small area? Which would you use to plan a long drive? Explain.

Extra Practice 5: Similar Triangles

Matching angles are marked with the same number of arcs. Arrow marks (>) show parallel lines. Diagrams are not drawn to scale.

A. Skills

1. Find the missing angle in each triangle. Are the two triangles similar?

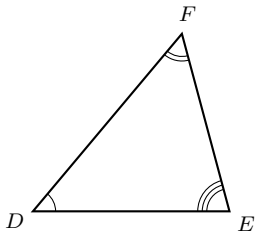
	Triangle 1 has	**Missing angle**	**Triangle 2 has**	**Missing angle**	**Similar?**
(a)	33° and 90°		57° and 90°		
(b)	48° and 66°		66° and 76°		
(c)	118° and 22°		40° and 22°		

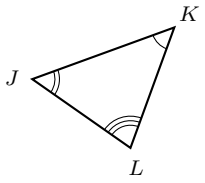
2. $\triangle UVW \sim \triangle XYZ$. Fill in the blanks.

(a) $\angle U = \angle$_____ (b) $\angle V = \angle$_____ (c) $\angle W = \angle$_____

(d) $\dfrac{XY}{UV} = \dfrac{_____}{VW} = \dfrac{XZ}{_____}$ (e) Side UV is across from $\angle W$. It matches side _____.

3. The two triangles are similar.





(a) Complete: $\triangle DEF \sim \triangle$_____

(b) Which side matches EF? _____

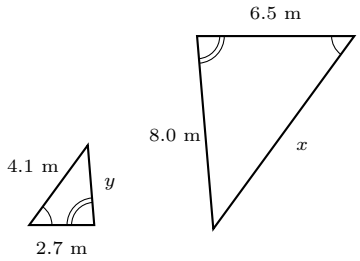
4. $\triangle PQR \sim \triangle XYZ$. Matching sides are in the same column.

$\triangle PQR$	$PQ = 5$ cm	$QR = 8$ cm	$PR = 11$ cm
$\triangle XYZ$	$XY = 7.5$ cm	$YZ =$	$XZ =$

Find the scale factor, then complete the table.

5. The two triangles are similar. Round to the nearest tenth of a metre.

(a) Find x. (b) Find y.

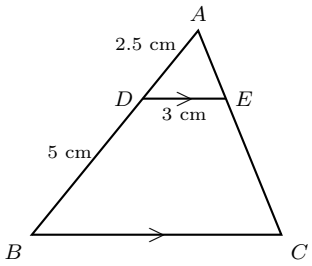


B. Applying

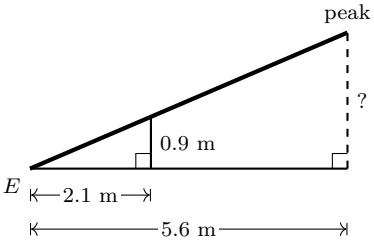
6. In $\triangle ABC$, $DE \parallel BC$. $AD = 2.5$ cm, $DB = 5$ cm and $DE = 3$ cm.

(a) Explain why $\triangle ADE \sim \triangle ABC$.

(b) Find BC.



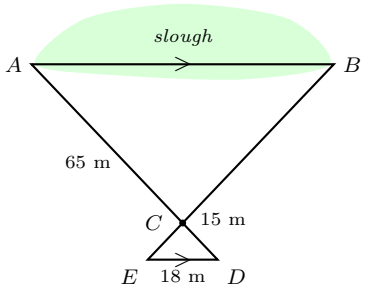
7. A roof rises steadily from the eave E to the peak. A vertical brace inside the roof is 0.9 m tall and stands 2.1 m (measured level) from the eave. The peak is 5.6 m from the eave, measured level. How tall is the roof at the peak?



8. A rancher wants to know the distance AB across a slough. He puts a stake at C. Points A, C, D are in a line, and so are B, C, E. He sets DE parallel to AB, then measures $AC = 65$ m, $CD = 15$ m and $DE = 18$ m.

(a) Give two reasons why $\triangle ABC \sim \triangle DEC$.

(b) How far is it across the slough?



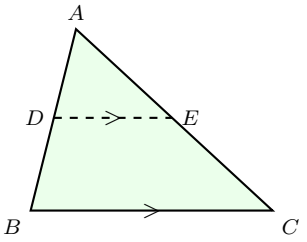
9. Two wheelchair ramps are built at the **same angle** with the level ground. Ramp 1 is 3.6 m long (along its surface) and rises 0.30 m. Ramp 2 must rise 0.45 m. How long must ramp 2 be? Explain why the two triangles are similar.

10. Taylor has two triangles. Triangle 1 has sides 3 cm, 4 cm and 5 cm. Triangle 2 has sides 6 cm, 8 cm and 9 cm. Taylor says, "Two of the sides doubled, so the triangles are similar." Is Taylor right? Show your work.

11. $\triangle PQR \sim \triangle XYZ$. $\angle P = 52°$ and $\angle Q = 61°$. What is the size of $\angle Z$?

A. 52° **B.** 61° **C.** 67° **D.** 113°

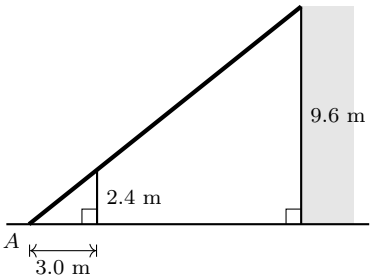
12. Numerical Response A garden bed is the triangle ABC. A straight path DE crosses it, parallel to side BC. $AD = 4.4$ m, $AB = 9.0$ m and $BC = 6.5$ m. The length of the path DE, to the nearest tenth of a metre, is _____ m.



C. Stretch

13. In $\triangle ABC$, $DE \parallel BC$ (D on AB, E on AC). $AD = 4$ cm, $DE = 5$ cm and $BC = 8.75$ cm. Find DB.

14. A straight grain auger runs from point A on the ground up to the top of a bin wall that is 9.6 m high. A vertical support post under the auger is 2.4 m tall and stands 3.0 m from A along the ground.



(a) How far from A is the bin wall, along the ground?

(b) A second post will stand 6.5 m from A. How tall must it be?

15. Explain why **any** two right triangles that each have a 25° angle must be similar. Then sketch two of them, of different sizes, and write all three angles in each.

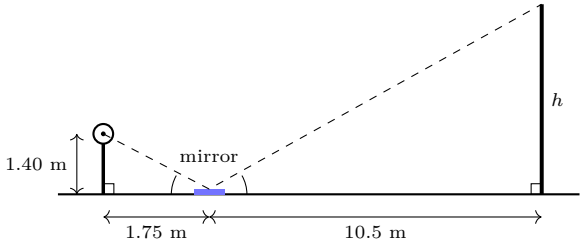
Extra Practice 6: Indirect Measurement: Shadows and Mirrors

A. Skills

1. In each row, the two shadows were measured at the same time. Find the missing length. Show your proportion below the table.

	Known object	**Its shadow**	**Other object**	**Its shadow**
(a)	stick: 1.0 m	1.25 m	flagpole:	9.0 m
(b)	post: 0.9 m	0.6 m	silo:	14.4 m
(c)	person: 1.8 m	2.7 m	sign: 4.2 m	

2. Use the mirror method to find the height h of the light pole.



3. Why must the two shadows in the shadow method be measured at the same time?

4. Ali holds a ruler upright 50 cm from his eye. A grain bin 150 m away covers 8 cm of the ruler, from its base to its top. How tall is the grain bin?

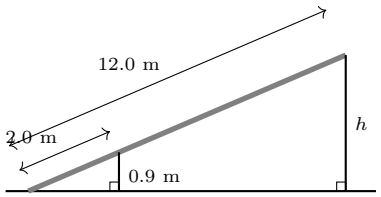
B. Applying

5. A yardstick (3 ft) stands upright and casts a shadow 2 ft 3 in long. At the same time, a tree casts a shadow 33 ft long. How tall is the tree?

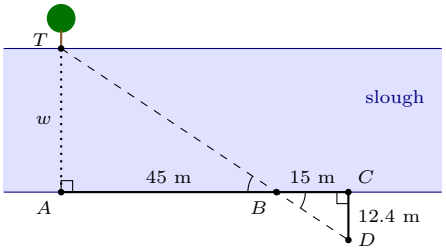
6. A scoreboard at a rodeo grounds is 7.2 m tall. Kayla puts a mirror on the ground 9.0 m from the bottom of the scoreboard. Her eyes are 1.62 m above the ground. How far from the mirror must she stand to see the top of the

scoreboard? Round to the nearest tenth of a metre.

7. A grain auger is a long straight tube that slopes up from the ground to the top of a bin. Its bottom end is on the ground. At a point 2.0 m along the auger from the bottom end, the auger is 0.9 m above the ground. The auger is 12.0 m long. How high is its top end?



8. Noah uses the stake method to find the width w of a slough (a prairie pond). He measures $AB = 45$ m, $BC = 15$ m and $CD = 12.4$ m. How wide is the slough?



9. Liam's eyes are 1.5 m above the ground. He stands 2.5 m from a mirror, and the mirror is 20 m from a building. He writes $h = 2.5 \times 20 \div 1.5 = 33.3$ m.

(a) What did Liam do wrong?

(b) Find the correct height. Explain why it is more reasonable.

10. At 6:00 p.m., a 2 m post casts a shadow 5 m long. Which statement is true at that moment?

A. Every shadow is 2.5 times as long as its object.

B. Every object is 2.5 times as tall as its shadow.

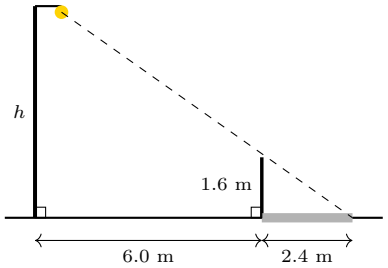
C. A 10 m tree has a shadow 4 m long.

D. Every shadow is 3 m longer than its object.

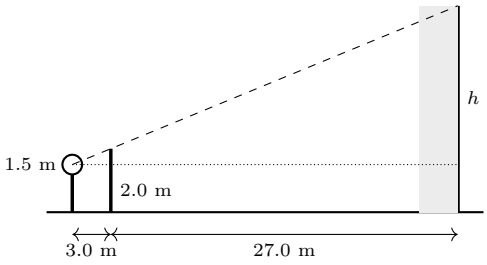
11. Numerical Response A 1.20 m stick casts a shadow 0.75 m long. At the same time, an apartment building casts a shadow 13.5 m long. The height of the building, to the nearest tenth of a metre, is _____ m.

C. Stretch

12. At night, Zoe (1.6 m tall) stands 6.0 m from a street lamp. The light from the lamp makes her shadow 2.4 m long. How high is the lamp? (Hint: the big triangle's base is the whole distance from the lamp post to the tip of her shadow.)



13. Owen's eyes are 1.5 m above the ground. He stands 3.0 m from a 2.0 m pole, so that the top of the pole lines up with the top of a building. The pole is 27.0 m from the building. How tall is the building?
(Hint: use only the parts *above* Owen's eye level, then add his eye height back on.)



14. Plan how you would find the height of your school using one method from this lesson. Name the tools you need, list the measurements you would take, draw a labelled sketch, and write the proportion you would solve. Give one thing that could make your answer a little off.

Extra Practice 7: Unit 6 Review

Practice Test 2. Treat this like the unit test. Show your work, and write each proportion before you solve it. Diagrams are not to scale unless the question says so.

A. Skills

1. Fill in the table. (k = new length ÷ original length.)

Original length	New length	k	Type
6 cm	4.2 cm		
1.8 m	5.4 m		
3.5 cm		2	
	4 cm	0.25	

2. $\triangle RST \sim \triangle UVW$. In $\triangle RST$, $RS = 8$ cm and $ST = 10$ cm. In $\triangle UVW$, $UV = 6$ cm and $UW = 9$ cm.

(a) List the three pairs of corresponding sides.

(b) Find the scale factor from $\triangle RST$ to $\triangle UVW$. Is it an enlargement or a reduction?

(c) Find VW and RT.

3. Three parallelograms are described in the table.

Parallelogram	Sides	Angles
P	4 cm and 6 cm	65° and 115°
Q	6 cm and 9 cm	65° and 115°
R	6 cm and 9 cm	70° and 110°

(a) Is P similar to Q? Show both tests.

(b) Is P similar to R? Explain.

(c) Are any two stop signs (regular octagons) similar? Explain.

4. A park map has a scale of $1 : 30\,000$.

(a) Write the scale in words, using metres.

(b) A bike path is 3.5 cm long on the map. How long is it really?

(c) A pond is really 1.8 km long. How long is it on the map?

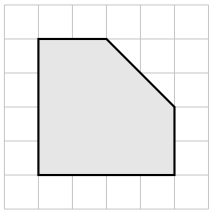
5. In $\triangle ABC$, $\angle A = 34°$ and $\angle B = 71°$. In $\triangle DEF$, $\angle D = 75°$ and $\angle E = 71°$.

(a) Are the triangles similar? Explain.

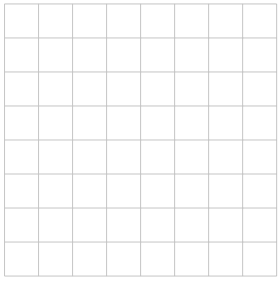
(b) Name them in matching order: $\triangle ABC \sim \triangle$_____

B. Applying

6. Enlarge the shape with a scale factor of 1.5. Draw it on the grid on the right.



original



$k = 1.5$

7. A school banner is 1.2 m by 2 m. The team wants a small copy for each player, using a scale factor of 0.35.

(a) Find the size of the small copy in centimetres.

(b) Check that the copy is similar to the banner.

8. Dana is drawing a plan of a deck that is 4.8 m long and 3.6 m wide. The space on her page is 10 cm by 8 cm.

(a) Which scale fits best: $1 : 40$, $1 : 50$ or $1 : 100$? Explain.

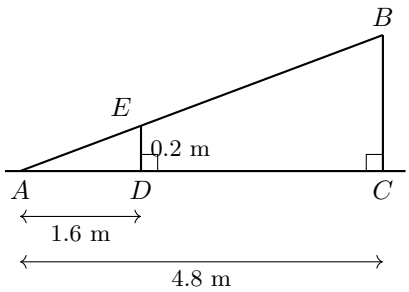
(b) The stairs are 1.2 m wide. How wide are they on her plan?

9. Before cutting down a dead tree, a worker wants to know if it could hit a shed 15 m away. The worker is 1.75 m tall and casts a 1.25 m shadow. At the same time, the tree casts a 10 m shadow.

(a) How tall is the tree?

(b) Can the tree reach the shed if it falls toward it? Why might the worker still want more room?

10. A loading ramp for a lawn tractor rises from the ground at A to a trailer deck at B. A support post DE stands straight up, 1.6 m from A, and is 0.2 m tall. Point C is on the ground right below B, 4.8 m from A.



(a) Explain why $\triangle ADE \sim \triangle ACB$.

(b) How high is the trailer deck, BC?

11. A 5 in by 8 in photo is enlarged to a poster 2.5 ft by 4 ft. The scale factor is

A. 0.5 **B.** $\frac{1}{6}$ **C.** 6 **D.** 25

12. Numerical Response A model car is built at a scale of 1 : 24. The real car is 4.56 m long. The model is _____ cm long.

C. Stretch

13. A building is 18 m tall. Erin puts a mirror on the ground 24 m from the building. Her eyes are 1.65 m above the ground. How far from the mirror must she stand to see the top of the building in it?

14. A floor plan has a scale of 1 : 80. A bedroom is 5 cm by 4 cm on the plan.

(a) Find the real length and width of the room, and its real area in square metres.

(b) The area on the plan is $5 \times 4 = 20$ cm^2. How many times bigger is the real area? Compare with 80×80.

15. Ken is 1.8 m tall and casts a 1.2 m shadow. A flagpole's shadow is 6 m. Ken writes $\frac{h}{1.8} = \frac{1.2}{6}$ and gets $h = 0.36$ m. Find his mistake and fix it.